

# DIVANSHU SHARMA

## MACHINE LEARNING SKILLS

|                         |                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| Languages & Framework:  | Python, Java, C++, SQL, TensorFlow, Keras, PyTorch, PyG.                                                 |
| Modeling & Algorithms:  | RNN, LSTM, GNN, CNN, Ensemble Methods (RF/XGBoost), Cross-Validation, Time Series, NLP, Transformers.    |
| Data Tools & Libraries: | Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly.                                                |
| Vector Search & LLMs:   | PubMedBERT, BioMistral-7B, Gemma, DeepSeek, Llama, Qdrant, Chroma, Retrieval-Augmented Generation (RAG). |

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github.com/sdivyanshu90

kaggle.com/divanshu22

CodeChef

## EDUCATION

|             |                                                                                                                                                                     |                    |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2021 - 2025 | <b>University of Mumbai</b><br>Bachelor in Engineering Computer Science   CGPA: 8.8<br>Relevant Coursework: Machine Learning, Deep Learning, Algorithms, Statistics | Computer Science   |
| 2022        | <b>Qiskit Global Summer School</b><br>Grade: Quantum Excellence                                                                                                     | Quantum Simulation |

## WORK EXPERIENCE

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 2022 - Present | <b>Research Consultant - WorldQuant BRAIN</b><br>Developed and backtested <b>100+</b> quantitative Alpha models using technical indicators and sentiment-based signals for equities in <b>US, Europe, and Asia</b> .<br>Engineered " <b>SuperAlphas</b> ," an ensemble approach that boosted portfolio Sharpe ratio by 8-10% across market regimes.<br>Backtested models on <b>10+</b> years of tick-level financial data, applying walk-forward validation to reduce overfitting. |
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## PROJECTS

|                              |                                                                                                                                                                                                                                                                                                                                                                                |
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| RAG, LLM, AI                 | <b>BioMed-Retrieval – Biomedical Q&amp;A using RAG &amp; LLMs</b><br>Architected a domain-specific Retrieval-Augmented Generation (RAG) system combining <b>PubMedBERT</b> , <b>Qdrant</b> , and <b>BioMistral-7B</b> .<br>Fine-tuned <b>BioMistral-7B</b> on biomedical Q&A pairs, improving relevance by 30% in offline evals; implemented hybrid semantic + keyword search. |
| Tensorflow, Keras, RNN, LSTM | <b>Next Word Prediction</b><br>Trained a <b>multi-layer LSTM network</b> on a <b>580K-sentence corpus</b> , achieving 92% next-token accuracy and reducing perplexity by 35%.                                                                                                                                                                                                  |
| GNN, PyTorch Geometric       | <b>Movies Recommendation - GNN and PyG</b><br>Implemented a <b>Graph Neural Network</b> recommender yielding <b>0.9282</b> validation AUC, outperforming matrix-factorization baseline by 15%.                                                                                                                                                                                 |
| Python                       | <b>MindfulML</b><br>Curated a GitHub repository with <b>140+</b> machine learning projects, covering diverse applications such as sentiment analysis, chatbot development, and stock price prediction.                                                                                                                                                                         |

## CERTIFICATIONS

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|------|------------------------------------------------------------------------------------------------------|----------|
| 2024 | <b>Generative Adversarial Networks (GANs) Specialization</b><br>Instructor: Sharon Zhou              | Coursera |
| 2023 | <b>Machine Learning Specialization</b><br>Instructor: Andrew Ng                                      | Coursera |
| 2023 | <b>Deep Learning Specialization</b><br>Instructor: Andrew Ng                                         | Coursera |
| 2023 | <b>DeepLearning.AI TensorFlow Developer Professional Certificate</b><br>Instructor: Laurence Moroney | Coursera |
| 2023 | <b>TensorFlow: Advanced Techniques Specialization</b><br>Instructor: Laurence Moroney                | Coursera |

## ACHIEVEMENTS

|               |                                                                                                                                                                                           |                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Quant Finance | <b>WorldQuant Brain</b><br>Achieved <b>GOLD LEVEL</b> recognition in the prestigious WorldQuant Challenge, demonstrating exceptional quantitative analysis and financial modeling skills. | Global Alphathon 2022                   |
| Quant Finance | <b>WorldQuant Brain</b><br>In recognition of ranking in the <b>top 5%</b> of <b>STAGE 1</b> of the <b>Global Alphathon 2022</b>                                                           | Global Alphathon 2022                   |
| Python        | <b>CodeChef</b><br>Solved <b>1200+</b> algorithmic problems using Python, demonstrating proficiency in data structures and computational techniques.                                      | Competitive Programming                 |
| API           | <b>Postman</b><br>Designed, tested, and documented 20+ RESTful endpoints using Postman collections                                                                                        | Postman API Fundamentals Student Expert |

## LANGUAGES

|         |                               |
|---------|-------------------------------|
| English | Full Professional Proficiency |
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